

Well Completion Report: Offset Incident Well MN-DW-02 (illustrative)

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1. Executive Summary

Drilled in Mahanadi deepwater block 1.8 km east of MN-SM-DW-01. At depth 4,195 m MD, while drilling ahead with 11.15 ppg SOBM, the well encountered an underbalanced Miocene turbidite channel sand resulting in a **12 bbl gas influx** with 236 psi SIDPP.

2. Influx Event Analysis

- Casing Shoe**: 13-3/8 in shoe at 3,840 m MD (FIT 12.05 ppg).
- Encountered Condition**: Normal compaction trend baseline departed at 4,152 m MD. Resistivity declined from 2.8 ohm-m to 1.4 ohm-m; compressional sonic increased from 86 us/ft to 104 us/ft.
- Kick Mechanism**: Mud weight was kept at initial 11.15 ppg. Top Miocene sand (U3) pressure was 11.48 ppg, creating a 0.33 ppg (150 psi) underbalance.
- Well Kill Execution**: The well was shut in with the subsea annular preventer. Wait & Weight kill method executed using **11.65 ppg (1.40 SG)** kill mud. 36 hours of NPT incurred.

3. Key Operational Recommendations

- Mandatory Pre-emptive Weight-Up**: For future deepwater wells in this fault compartment, mud weight must be raised to 11.65 ppg *before* penetrating 4,195 m MD.
- ROP Restriction**: Impose 11–12 m/hr ROP restriction across the sand boundary.